The build is the easy half. Trust is the product.

A realistic support flow, the controls around it, and the LangSmith loop that caught a fluent failure no error dashboard could see.

THE RELIABILITY PROBLEM

"I'm handing this to Steve… I've passed along your concern."

Tool calls: none

Warm tone. Correct facts. Green trace. The customer waits for help that was never requested.

BUILT HERE

LangChain

Fast, configurable agent harness. Six tools and middleware on one standard loop.

RUNTIME UNDERNEATH

LangGraph

Durable state, explicit topology, streaming, checkpointing, and human interrupts.

GRADUATE WHEN NEEDED

Deep Agents

Long-horizon planning, subagents, skills, and a virtual filesystem.

LangSmith — Observe — Evaluate — Improve

Business hypothesis — resolve billing accurately, then use the resolved moment for discovery. Recommendation attach rate is unknown until real traffic.

100M+ monthly OSS downloads **6K+** active LangSmith customers **5 of Fortune 10** **35%** of Fortune 500 **$125M Series B** at $1.25B

Sources: LangChain · verified 05 Aug 2026

One rule: Helena can only act as Helena.

The model understands the request and chooses a tool. Code fixes the customer identity, scopes the data, calculates money, and controls writes.

SIGN IN

Verify who is calling

Helena's bearer credential resolves to customer 6. The request body cannot claim customer 26.

THREAD

Protect the saved conversation

Only Helena can open or resume Helena's thread—before its history is loaded.

DATA

Scope every read

Tools expose no customer picker; parameterized SQL binds Helena's ID and computes totals.

ACTION

Control side effects

Refunds pause for approval; stable keys prevent a retry from creating a duplicate ticket.

FLAT CREATE_AGENT · SIX TOOLS

Model: language + tool choice

BILLING · 3	List invoices · invoice detail · SQL spend summary
DISCOVERY · 1	Catalog-grounded recommendations excluding owned tracks
SUPPORT · 2	File refund request · queue human handoff

Model never chooses
Customer · SQL · billing totals

Code enforces
Approval · idempotency · audit guard

TWO ATTACKS, KEPT SEPARATE

Thread attack: Richard → Helena's thread. Blocked before conversation history loads.

Identity attack: Helena credential + Richard's ID. Rejected before a run starts.

Implementation: src/agent/graph.py:86 src/data/db.py:227 src/agent/tools/support.py:53 Studio identity is a developer-side simulation, not production auth. src/security/auth.py:190

Complexity did not earn its hop. Flat ships.

Same model, same 30-example dataset, same evaluators, same commit. The supervisor had to clear every pre-registered bar.

SHIPS

Flat

Resolved	30 / 30
p50 full turn	3.8s
Tool calls	45
Security failures	0

CHALLENGER

Supervisor

Resolved	30 / 30
p50 full turn	6.6s
Tool calls	61
Security failures	0

WHAT MADE THE DECISION CREDIBLE

The first controlled freeze showed 29–30. I did not bank a one-example lead.

The bar was written before the challenger existed. After the escalation fix, both arms were rerun at the same commit and tied. What survived was the extra hop: 1.7× median latency and 16 more tool calls.

Decision rule: visible quality gain + no workflow regression + ≤2.0s latency penalty. Ties go to the simpler topology.

PROVENANCE (BOTH ARMS) commit a29a530 · clean dataset 6ba144216f8d evaluators decda2664959

Measured, not inferred. 30 examples are a decision set, not a statistical claim.

Artifact: reports/experiment.md

A trace proves a call happened. Not that the promise is true.

Core message: a believable answer is not enough. I move from a real failure to a replayable test, a coded quality rule, and ongoing monitoring.

01 · FLUENT CLAIM

"I'm handing this to Steve."

```
tool calls: none
side effect: none
```

→

02 · TOOL APPEARS

Escalation tool returns prose

```
tool call: yes
durable row: none
```

→

03 · EFFECT EXISTS

Row queued; reply says "passed"

```
row: queued
assigned / sent: no
```

→

04 · HONEST CLAIM

"Queued; not assigned or sent."

```
row: queued
claim matches state
```

Trace
find failure
Production
monitor samples
›
Replay
reproduce state
›
Dataset
freeze case
›
Evaluator
encode rule
›
Experiment
prove fix
›

State-aware replay — clone a traced thread back into the local graph.

Trace → dataset — turn a real failure into a permanent regression case.

One quality definition — reuse evaluators offline and online.

Code for truth: exact facts, authorization, writes, trajectories **Judges for judgment:** tone and helpfulness

No deployment features shown.

Pilot two workflows. Stop on two failures.

The build is ready for a bounded evaluation—not a production victory lap. The missing numbers require real customers and real reviewers.

Measured on this build

Task resolution, frozen comparison	30 / 30
Cross-tenant access / unapproved refund writes	0 / 0
Flat p50 full-turn latency	3.8s
Deterministic pre-flight	139 passed

Unknown until production traffic

Containment without unsafe resolution	unknown
Recommendation attach / conversion rate	unknown
Human review agreement and downstream outcome	unknown
Actual queue shape and integration effort	unknown

FRICTION WORTH FIXING

- Authorization payload hid run context under `kwargs`; the obvious check failed open.
- Useful HITL cards required async prefetch because the description callback cannot do I/O.
- Middleware composed a prompt no UI exposed—the root cause of the fabricated handoff.

TWO-WEEK PILOT

- **Scope** — billing lookup + refund intake
- **Measure** — containment + unsafe-resolution
- **Safety stop** — any foreign-tenant access or unapproved refund write

Next: add support-rep annotation queue; hold recommendations behind an attach-rate experiment.

The build is not the bottleneck. Getting to reliable is.